

HACKATHON ALPINE CLIMATE DATA CHALLENGE: TASK AND DATA SOURCE

Task

1. Local climate model

This task aims to develop a customized climate model for the two valleys (Susa Valley and Maurienne Valley), combining historical climate data with IPCC scenarios. It is essential to simulate the evolution of key climate variables, such as temperature, precipitation, wind, humidity, atmospheric pressure and snow cover, to estimate the future impact of climate change on the region. The output must be made available through the creation of interactive tools and visual content to facilitate understanding and dissemination of the results.

The activity should include:

- I. Cleaning and normalization of climate data to ensure temporal and spatial consistency
- II. Identification of recurring patterns and seasonal changes
- III. Analysis of local topographic and climatic characteristics to calibrate the model
- IV. Integration of global and regional datasets (e.g. ERA5, Copernicus, etc.) to obtain a complete and accurate understanding
- V. Use of dynamic climate models (e.g. WRF, etc.) to simulate future scenarios up to 2050
- VI. Validation of the model through comparison with observed climate data
- VII. Development of a dashboard structure that centralizes the simulated climate data, allowing an intuitive understanding of the predicted scenarios

Expected output:

- ❖ **Local climate model:** Detailed simulations up to 2050, with emphasis on seasonal changes and extreme conditions
- ❖ **Interactive dashboard:** Exploration of the main local climate scenarios and comparison with global trends
- ❖ **App prototype:** Interactive and educational tool to raise awareness on the potential impacts of climate change in schools, institutions and local communities
- ❖ **Advanced UI/UX components:** Intuitive layouts, thematic maps and narrative content to make the impact of climate change at a regional level more understandable, offering an intuitive user experience, suitable for both experts and the general public

Data source

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